SAPIR-Net

Supply Chain Analysis for Pharmaceutical Infrastructure Resilience

A Network Model for Quantifying Cascading Vulnerability in the U.S. Generic Oncology Drug Supply Chain

Phase 1: Cisplatin, Carboplatin, and Methotrexate

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1. Executive Summary and Introduction

1.1 Executive Summary

The United States lacks a quantitative framework for assessing geopolitical concentration risk in the upstream supply chains of essential generic oncology drugs. This paper presents SAPIR-Net (Supply Chain Analysis for Pharmaceutical Infrastructure Resilience — Network Model), a three-layer directed graph model that maps the flow of chemical precursors from geopolitical source nations through intermediate commodity nodes to three essential cancer therapies: cisplatin, carboplatin, and methotrexate. The model integrates empirical bilateral trade data from the United Nations Comtrade database (2019–2023), chemical dependency mappings, and stochastic disruption simulations to quantify the probability of severe drug shortages under defined geopolitical and logistical scenarios.

Three principal findings emerge from the analysis.

First, Herfindahl-Hirschman Index (HHI) scores for both platinum-group compounds (HS 284390) and heterocyclic nitrogen compounds (HS 293359) imported by the United States exceed 9,000 on a 10,000-point scale across all five years studied. By the standard thresholds established by the Department of Justice and Federal Trade Commission, both commodity flows are classified as highly concentrated. However, the concentration is dominated by a residual "Rest of World" category — not by China or India directly — exposing a structural limitation in 6-digit Harmonized System trade data that obscures the true distribution of pharmaceutical-grade precursor flows within broader industrial commodity categories. This paper terms this phenomenon the "6-Digit Dilution Effect" and identifies it as a critical blind spot in current federal supply chain monitoring.

Second, Monte Carlo simulation (N = 10,000 iterations per scenario) reveals a sharp divergence in vulnerability profiles across the three drugs. Under a cascading upstream shock scenario — in which direct Chinese exports are eliminated and the Rest of World supply is probabilistically degraded to reflect documented indirect Chinese dependency in starting material supply chains — methotrexate faces a 100% probability of severe shortage (defined as greater than 30% capacity loss), with a mean capacity reduction of 59%. Cisplatin and carboplatin, by contrast, remain below the severe shortage threshold under the same scenario, with mean capacity losses of approximately 17%. This divergence is driven by the commodity-specific structure of upstream dependency: heterocyclic compound supply chains exhibit 50–80% indirect exposure to Chinese starting materials, while platinum-group compound supply chains, dominated by South African mining operations, exhibit only 10–30% indirect Chinese exposure.

Third, the model's baseline geopolitical disruption scenario (Scenario A: complete loss of direct Chinese and Indian exports) produces a predicted capacity loss of 2.95% for platinum-based drugs. This figure falls within the 95% confidence interval of the observed real-world impact of the 2023 cisplatin and carboplatin shortage (2.7% average reduction; 95% CI: −4.4% to −0.9%), as documented in a 2025 cohort study of 11,797 U.S. cancer patients published in the Journal of the National Cancer Institute (Reibel et al., 2025).

These findings indicate that a single-policy response to oncology drug supply chain risk is structurally inadequate. Methotrexate requires upstream chemical diversification away from Chinese intermediate suppliers. The platinum-based agents require logistics corridor redundancy and domestic manufacturing investment. Current federal monitoring tools, constrained by the 6-Digit Dilution Effect, cannot distinguish between these risk profiles without the type of commodity-specific network modeling presented here.

1.2 Introduction

1.2.1 The Problem

Drug shortages in the United States are a persistent and worsening failure of pharmaceutical infrastructure. At the end of 2024, the U.S. Food and Drug Administration reported 98 active and ongoing drug shortages (Malta et al., 2025). The problem is concentrated in a specific class of products: older, off-patent, generic sterile injectable drugs with low per-unit prices and limited manufacturer redundancy. The FDA Drug Shortages Task Force identified three root causes: lack of economic incentives to produce less profitable drugs, market failure to recognize and reward mature quality management systems, and logistical and regulatory barriers that prevent supply recovery after disruptions (FDA, 2019).

Oncology drugs occupy the most vulnerable position within this landscape. An analysis by the U.S. Pharmacopeia found that 7 of 20 essential cancer medicines assessed were three to five times more likely to experience shortage than the average prescription drug product (Malta et al., 2025). Essential injectable cancer drugs currently in shortage have an average wholesale price of $2 per vial; those not in shortage average $1,423 per vial. This 700-to-1 price differential creates a structural disincentive for manufacturers to invest in production redundancy, quality systems, or supply chain resilience for the drugs that patients most critically need.

Three generic oncology drugs — cisplatin, carboplatin, and methotrexate — illustrate this failure with particular clarity. All three are included on the World Health Organization's List of Essential Medicines. All three are sterile injectables. All three experienced concurrent shortages beginning in early 2023. As of April 2026, carboplatin and methotrexate remain in active shortage according to the FDA Drug Shortage Database, with multiple manufacturers reporting unavailability or discontinued production (FDA, 2026a; FDA, 2026b). Cisplatin's shortage, first posted February 10, 2023, was resolved in June 2024 but reappeared on the American Society of Health-System Pharmacists shortage list in May 2025 (Malta et al., 2025).

1.2.2 The Upstream Dimension

The existing literature on drug shortages focuses predominantly on downstream factors: manufacturer concentration, pricing dynamics, regulatory barriers to market entry, and hospital inventory management. Comparatively little quantitative work addresses the upstream geopolitical dimension — the concentration of raw material and chemical precursor supply chains in a small number of countries, and the cascading consequences of disruption at that level.

This gap is not for lack of recognition. The National Academies of Sciences, Engineering, and Medicine concluded in 2022 that "lack of transparency and limited visibility into medical product supply chains have led to limited empirical evidence for identifying best strategies to address supply chain issues" (NASEM, 2022). The NASEM committee's Recommendation 2 called for the establishment of a publicly accessible database to map supply chain vulnerabilities and perform risk assessments under both normal and emergency scenarios. The White House 100-Day Supply Chain Review under Executive Order 14017 similarly recommended that the Department of Health and Human Services develop new authorities to "track production by facility, track API sourcing, and require API and finished dosage form sources can be identified on labeling for all pharmaceuticals sold in the United States" (The White House, 2021). That review noted that generic drugs represent 90 percent of all pharmaceuticals consumed in the United States, yet "there is little transparency into the origins of API within generic drugs."

The data confirm the scale of the exposure. As of 2019, 87 percent of generic active pharmaceutical ingredient manufacturing facilities were located outside the United States, concentrated in India (26 percent) and China (18 percent) (Kaygisiz et al., 2019; confirmed by The White House, 2021). India, which is the largest provider of generic finished dosage form products to the U.S. market, depends on China for approximately 70 percent of its active pharmaceutical ingredients (NASEM, 2022, citing Schondelmeyer et al., 2020; independently corroborated by The White House, 2021, citing industry analysis; see also Huang, 2019). This creates a layered dependency structure in which the United States imports finished drugs from India, which imports APIs from China, which controls key starting materials — a chain that is largely invisible in standard bilateral trade statistics.

1.2.3 The 2023 Platinum Shortage as a Case Study

The consequences of this hidden upstream concentration became visible in February 2023, when the FDA announced a cisplatin shortage triggered by the shutdown of a single pharmaceutical manufacturing facility — Intas Pharmaceuticals Limited in Sanand, Gujarat, India. An FDA inspection in November 2022 identified systemic current good manufacturing practice violations including destruction of quality control documents, falsification of environmental monitoring data, hundreds of aborted chromatographic sequences, and failures of executive management oversight (FDA, 2023). The facility was placed on Import Alert 66-40 in June 2023, effectively blocking all products from entering the United States.

The resulting shortage was quantified in a retrospective cohort study of 11,797 adults with advanced solid cancers initiating first-line therapy across geographically diverse U.S. oncology clinics. During the year following the shortage announcement, platinum chemotherapy prescribing declined by 2.7 percentage points overall and by 15.1 percentage points at peak shortage in June 2023, corresponding to an estimated 137 fewer patients than expected receiving platinum-based treatment (Reibel et al., 2025). The FDA authorized emergency importation of cisplatin with non-U.S. labeling to mitigate the crisis.

The facility-level production data confirm the concentration risk. As of January 2025, one facility in India produced 24 percent of the U.S. cisplatin supply, 8 percent of carboplatin, 84 percent of temsirolimus, and 43 percent of carmustine (Malta et al., 2025, citing USP Medicine Supply Map data).

A single-site failure at such a node does not disrupt one drug — it disrupts multiple drugs simultaneously.

1.2.4 Scope and Objectives of This Paper

This paper presents SAPIR-Net, a macro-geopolitical dependency graph model designed to quantify upstream concentration risk for three essential U.S. oncology drugs. The model operates at the level of national commodity flows rather than proprietary facility-level data, using publicly available United Nations Comtrade bilateral trade statistics and validated chemical dependency mappings to construct a three-layer directed network. Stochastic disruption scenarios are propagated through the network using Monte Carlo simulation with empirically grounded distributional assumptions.

The objectives are threefold:

First, to quantify the geographic concentration of U.S. imports for two chemical precursor categories — platinum-group compounds (HS 284390) and heterocyclic nitrogen compounds (HS 293359) — and to identify the structural limitations of 6-digit Harmonized System classification in isolating pharmaceutical-grade trade flows from broader industrial commodity volumes.

Second, to model the cascading impact of upstream disruption scenarios on the downstream availability of cisplatin, carboplatin, and methotrexate, incorporating both direct bilateral trade dependencies and indirect upstream exposure through third-country supply chains.

Third, to compare the model's predictions against the observed impact of the 2023 platinum chemotherapy shortage to assess the model's validity as a tool for prospective supply chain risk assessment.

2. Methodology and Network Taxonomy

2.1 Overview

The SAPIR-Net model is structured as a three-layer directed graph (digraph) implemented in Python using the NetworkX library. Nodes represent geopolitical entities (Layer 1), chemical commodity classes (Layer 2), and essential medicines (Layer 3). Directed edges represent material flows between layers, weighted by empirical trade data (L1 → L2) and binary chemical dependency flags (L2 → L3). Disruption scenarios are applied to the graph by modifying edge weights according to scenario-specific rules, and the downstream impact on Layer 3 medicine nodes is computed by propagating the modified flows through the stoichiometric dependency structure.

The model is implemented as four independent, auditable modules: (1) graph construction and concentration analysis, (2) disruption probability engine, (3) Monte Carlo simulation loop, and (4) visualization and reporting. Each module accepts the prior module's output as input, with no hidden state dependencies between modules.

2.2 Data Sources

2.2.1 UN Comtrade Bilateral Trade Data

The primary empirical input is bilateral trade data extracted from the United Nations Comtrade database (comtradeplus.un.org) for calendar years 2019 through 2023. The United States (Comtrade reporter code 842) is the importing country in all records. Three partner groups are represented: China (code 156), India (code 699), and World (code 0, representing the global aggregate). The World aggregate is used to derive a "Rest of World" residual by subtracting the China and India bilateral values from the World total.

Two Harmonized System 6-digit commodity codes are targeted:

HS 284390 — "Inorganic or organic compounds of precious metals, not elsewhere classified; amalgams of precious metals." This category includes platinum-group metal compounds used as precursors in the synthesis of platinum-based chemotherapy agents (cisplatin, carboplatin). It also includes non-pharmaceutical industrial applications (automotive catalytic converter compounds, electronics materials), which constitute the majority of trade volume at the 6-digit level.

HS 293359 — "Heterocyclic compounds containing a pyrimidine ring or piperazine ring in the structure, not elsewhere classified." This category includes intermediates used in the synthesis of methotrexate and other folate antagonist drugs. It also encompasses a wide range of non-pharmaceutical heterocyclic compounds used in agrochemistry and industrial chemistry.

The inclusion of non-pharmaceutical trade volumes within these 6-digit codes is a known limitation, discussed in Section 2.6 as the "6-Digit Dilution Effect."

2.2.2 Chemical Dependency Mappings

The dependency relationships between Layer 2 commodity nodes and Layer 3 medicine nodes are derived from established pharmaceutical chemistry. Cisplatin (cis-diamminedichloroplatinum(II)) and carboplatin (cis-diammine(1,1-cyclobutanedicarboxylato)platinum(II)) both require platinum-group metal compounds as obligate precursors. Methotrexate (4-amino-10-methylfolic acid) requires heterocyclic nitrogen-containing intermediates. These dependencies are binary: synthesis is not possible without the precursor. No continuous weight is assigned to these edges.

2.2.3 Upstream Exposure Parameters

Indirect Chinese dependency within the Rest of World supply chain is parameterized using the variable row_china_exposure, defined as the estimated fraction of Rest of World trade flow that ultimately depends on Chinese upstream inputs (starting materials, intermediates, bulk chemicals). This parameter is commodity-specific:

HS 284390 (Platinum compounds): 0.10–0.30. Justification: Global platinum mine production is dominated by South Africa, which produced 120,000 kg of the 180,000 kg global total in 2023 (66.7%). China does not appear as a named producer in USGS statistics, falling within the "Other countries" category at approximately 2.6% of global output. China is also absent from the named U.S. import sources for platinum (South Africa 33%, Switzerland 15%, Germany 14%, Belgium 9%) (Schulte, 2024). Indirect Chinese exposure for platinum compounds is therefore limited to secondary refining and specialty compound synthesis, supporting a low exposure band.

HS 293359 (Heterocyclic compounds): 0.50–0.80. Justification: India, the largest provider of generic finished dosage forms to the U.S. market, depends on China for approximately 70 percent of its active pharmaceutical ingredients (NASEM, 2022; The White House, 2021). European and Japanese API manufacturers similarly rely on Chinese bulk intermediate chemicals for heterocyclic compound synthesis. The White House 100-Day Supply Chain Review under E.O. 14017 confirmed that "China and India are estimated to control substantial parts of the supply chain" for pharmaceutical products, while acknowledging that "lack of data and supply chain transparency make it difficult to estimate the precise share" (The White House, 2021). The 0.50–0.80 band reflects the convergent evidence from three independent sources: NASEM (2022, citing Schondelmeyer et al., 2020), the White House (2021, citing industry analysis), and Huang (2019, cited by the White House).

2.3 Node Taxonomy

Layer 1: Geopolitical Source Nodes

Node	Description	Role in Model
CN	China	Direct exporter of precursor chemicals to the U.S.; upstream starting material supplier to Indian and European API manufacturers
IN	India	Direct exporter of precursor chemicals and finished dosage forms to the U.S.; dependent on Chinese upstream inputs
ROW	Rest of World	Residual aggregate (World minus CN minus IN); includes EU, Japan, South Africa, and other exporters

Layer 2: Chemical Chokepoint Nodes

Node	HS Code	Description	Downstream Dependencies
PLAT	284390	Platinum-group compounds	Cisplatin, Carboplatin
HETERO	293359	Heterocyclic nitrogen compounds	Methotrexate

Layer 3: Essential Medicine Nodes

Node	Drug	Chemical Class	FDA Shortage Status (April 2026)
CIS	Cisplatin	Platinum-based alkylating agent	Resolved Jun 2024; recurred May 2025 (ASHP)
CARB	Carboplatin	Platinum-based alkylating agent	Active shortage since Apr 28, 2023
MTX	Methotrexate	Folate antagonist	Active shortage since Mar 13, 2023

2.4 Edge Weighting

2.4.1 L1 → L2 Edges: Empirical Trade Flows

Edges connecting geopolitical source nodes to chemical commodity nodes are weighted by five-year aggregate trade values (USD) extracted from the UN Comtrade dataset (2019–2023). Net weight in kilograms is retained as a secondary variable. The five-year aggregation smooths annual volatility and provides a more stable representation of structural trade relationships.

2.4.2 L1 → L2 Concentration Metric: Herfindahl-Hirschman Index

Geographic concentration of each commodity's import flow is evaluated using the Herfindahl-Hirschman Index, computed annually per commodity:

$$HHI = \Sigma(s_i)^2$$

where s_i is the market share (as a percentage of total U.S. import value) of source country i. Under DOJ/FTC guidelines, an HHI below 1,500 indicates an unconcentrated market, 1,500–2,500 indicates moderate concentration, and above 2,500 indicates high concentration. An HHI of 10,000 represents a single-source monopoly.

2.4.3 L2 → L3 Edges: Binary Stoichiometric Flags

Edges connecting chemical commodity nodes to essential medicine nodes carry binary dependency flags (1.0 or 0.0) based on chemical necessity. A flag of 1.0 indicates that the upstream commodity is an obligate precursor for the downstream drug — synthesis is not chemically possible without it. A flag of 0.0 indicates no chemical dependency; no edge is created.

Commodity	Drug	Flag	Rationale

PLAT (HS 284390)	Cisplatin	1.0	Platinum atom required in molecular structure
PLAT (HS 284390)	Carboplatin	1.0	Platinum atom required in molecular structure
PLAT (HS 284390)	Methotrexate	0.0	No chemical dependency (no edge)
HETERO (HS 293359)	Methotrexate	1.0	Heterocyclic intermediate required for synthesis
HETERO (HS 293359)	Cisplatin	0.0	No chemical dependency (no edge)
HETERO (HS 293359)	Carboplatin	0.0	No chemical dependency (no edge)

No continuous weights are assigned to L2 → L3 edges. The vulnerability quantification originates entirely from the empirical L1 → L2 layer. L2 → L3 edges define the structural propagation path for disruption — which drugs are affected when a given commodity flow is interrupted — not the magnitude of the disruption itself.

2.5 Disruption Scenarios and Stochastic Framework

Three disruption scenarios are defined, each representing a distinct class of supply chain threat. A baseline (no-shock) condition is included for reference.

Scenario A: Direct Export Ban

A deterministic scenario in which all direct exports from China and India to the United States for both commodity codes are reduced to zero. Rest of World flows are unaffected. This scenario models a complete bilateral trade disruption — such as an export ban, comprehensive tariff barrier, or simultaneous regulatory enforcement action — without accounting for indirect upstream dependencies.

Scenario B: Cascading Upstream Shock

China's direct exports to the United States are eliminated. Additionally, the Rest of World supply is probabilistically degraded to reflect the estimated fraction of ROW production that depends on Chinese upstream inputs. The degradation fraction is sampled independently for each commodity code from a rescaled Pareto distribution bounded by the commodity-specific row_china_exposure range.

The Pareto distribution is selected on empirical grounds: supply chain disruption severity exhibits fat-tailed behavior, with most events causing moderate degradation but a non-trivial probability mass allocated to near-total upstream collapse. The shape parameter $\alpha = 2.5$ produces a moderately heavy tail with finite variance. Under this parameterization, the bulk of sampled degradation values cluster near the lower bound of the exposure range, with the tail extending toward the upper bound.

For HS 284390 (platinum compounds), sampled ROW degradation centers at approximately 15.7% (median 14.8%), with a 95th-percentile value of 23.9%. For HS 293359 (heterocyclic compounds),

sampled ROW degradation centers at approximately 58.6% (median 57.3%), with a 95th-percentile value of 70.8%.

India's direct exports are not zeroed in Scenario B. This reflects the scenario's design as a China-specific upstream shock rather than a generalized bilateral disruption.

Scenario C: Global Logistics Chokepoint

A systemic capacity reduction is applied uniformly to all L1 → L2 edges, regardless of origin country. This scenario models non-geopolitical disruptions — major port closures, pandemic-era shipping constraints, catastrophic weather events — that degrade the entire import corridor simultaneously.

The capacity reduction fraction is sampled from a log-normal distribution with parameters calibrated to produce a median reduction of approximately 20% and a 95th-percentile reduction of approximately 46%. The log-normal distribution captures the right-skewed empirical profile of logistics disruptions: most events cause moderate capacity loss, but extreme events (Suez Canal blockage, COVID-era port shutdowns) can reach 50–70% reduction.

Simulation Parameters

Each scenario is executed for N = 10,000 independent iterations. In each iteration, the relevant disruption parameters are sampled from their specified distributions, applied to the graph's edge weights, and propagated through the L2 → L3 stoichiometric dependency structure. The resulting capacity loss for each Layer 3 medicine node is recorded. A "severe shortage event" is defined as a capacity loss exceeding 30%, consistent with FDA supply vulnerability thresholds.

Output metrics computed per drug per scenario include: mean capacity loss, median capacity loss, 5th and 95th percentile capacity loss, and the probability of a severe shortage event (proportion of iterations exceeding the 30% threshold).

2.6 Known Limitations

The 6-Digit Dilution Effect. The two HS codes used in this analysis (284390, 293359) encompass broad categories of chemical compounds, the majority of which are consumed in non-pharmaceutical applications. HS 284390 includes platinum compounds used in automotive catalytic converters, electronics, and industrial catalysis — volumes that substantially exceed pharmaceutical demand. HS 293359 includes heterocyclic compounds consumed in agrochemistry, dye manufacturing, and other industrial processes. As a result, the bilateral trade flows observed in Comtrade data reflect the aggregate industrial demand for these compounds, not the pharmaceutical-specific subset. The HHI scores computed from these data characterize the concentration of total compound imports, not the concentration of pharmaceutical-grade precursor imports specifically.

This limitation is structural to the Harmonized System classification at the 6-digit level. More granular national tariff codes (8-digit or 10-digit) may isolate pharmaceutical-grade subsets, but these are not standardized across countries and are not available in the Comtrade international reporting framework. The model therefore operates as a conservative upper bound on supply diversification: if the total commodity flow is highly concentrated, the pharmaceutical subset —

which represents a smaller and less commercially attractive fraction — is unlikely to be more diversified.

Parameterized indirect exposure. The row_china_exposure variable is an informed estimate, not an empirically measured trade flow. While the parameter ranges are grounded in convergent evidence from federal reports and peer-reviewed sources, the precise fraction of Rest of World pharmaceutical production that depends on Chinese starting materials is not directly observable in any publicly available dataset. The model addresses this limitation by presenting results as sensitivity bands across the full parameter range rather than as point estimates.

Binary stoichiometric model. The L2 → L3 dependency structure assumes zero inventory buffer, zero substitution elasticity, and instantaneous propagation of upstream supply deficits to downstream drug availability. In practice, manufacturers maintain variable safety stocks, hospitals may hold weeks-to-months of inventory, and partial substitution between drug regimens is clinically feasible for some patient populations. The model therefore represents a worst-case propagation scenario. The 2023 platinum shortage data suggest that real-world mitigation mechanisms (alternative regimens, emergency imports, dose rounding) attenuated the observed impact relative to what a zero-buffer model would predict — a finding consistent with the model's conservative design.

Static graph structure. The model does not account for dynamic market responses to disruption, such as new market entrants, capacity expansion by existing manufacturers, or trade route rebalancing. These responses operate on timescales of months to years and are not captured in a single-period simulation. The model is therefore best interpreted as a measure of acute vulnerability — the immediate-term exposure before market adjustment mechanisms can operate.

3. Simulation Results and Risk Divergence

3.1 Concentration Analysis

Herfindahl-Hirschman Index scores computed from UN Comtrade bilateral import data for the United States confirm that both precursor commodity flows are highly concentrated across all five years of the study period.

For HS 284390 (platinum-group compounds), the five-year average HHI is 9,415. Annual values range from 9,137 (2019) to 9,685 (2020). For HS 293359 (heterocyclic nitrogen compounds), the five-year average HHI is 9,114. Annual values range from 8,900 (2021) to 9,227 (2020). Both commodities exceed the DOJ/FTC high-concentration threshold of 2,500 by a factor of approximately 3.6.

The source of concentration, however, is not China or India. China's share of U.S. platinum compound imports (by trade value) ranges from 0.5% to 2.4% across the study period. India's share ranges from 0.6% to 2.3%. The Rest of World residual accounts for 94–98% of import value in every year. For heterocyclic compounds, China's share ranges from 2.4% to 3.6%, India's from 1.1% to 2.2%, with the Rest of World residual absorbing 94–96%.

These results expose what this paper terms the 6-Digit Dilution Effect. The dominant volumes in both HS codes are non-pharmaceutical industrial applications — catalytic converter compounds for HS 284390, agrochemical intermediates for HS 293359. The pharmaceutical-grade fraction of trade flow is a small subset of the aggregate, invisible at the 6-digit classification level. The high HHI scores reflect concentration of total industrial demand across a small number of advanced-economy exporters (primarily the European Union, Japan, and South Africa for platinum), not concentration of pharmaceutical precursor supply in China or India specifically.

This finding does not diminish the vulnerability — it relocates it. The direct bilateral China-to-U.S. trade flow for these commodities is small. The risk resides in the indirect dependency channel: Rest of World exporters (EU, Japan, India) that themselves depend on Chinese starting materials for pharmaceutical-grade synthesis. This indirect dependency is the mechanism modeled in Scenario B.

3.2 Monte Carlo Simulation Outputs

The Monte Carlo simulation was executed for N = 10,000 iterations per scenario across four conditions: Baseline (no disruption), Scenario A (direct export ban on China and India), Scenario B (cascading upstream shock with commodity-specific ROW degradation), and Scenario C (global logistics chokepoint). Results are reported for each Layer 3 medicine node.

3.2.1 Baseline

Under baseline conditions, all three drugs register zero capacity loss across all 10,000 iterations. This confirms the model's internal consistency: absent disruption, the full trade flow reaches each medicine node without attenuation.

3.2.2 Scenario A: Direct Export Ban

Eliminating all direct Chinese and Indian exports to the United States produces a uniform capacity loss of 2.95% for cisplatin and carboplatin and 4.58% for methotrexate across all iterations. Because this scenario is deterministic — no stochastic component — the loss is identical in every iteration. Neither drug breaches the 30% severe shortage threshold. The probability of severe shortage is zero for all three drugs.

This result reflects the small direct bilateral trade share that China and India hold in these commodity codes. Removing their contribution leaves the Rest of World supply channel — which carries 94–98% of the trade volume — entirely intact. Under a purely bilateral framing, the U.S. oncology drug supply chain appears resilient to geopolitical disruption. That appearance is misleading, as Scenario B demonstrates.

3.2.3 Scenario B: Cascading Upstream Shock

Scenario B eliminates China's direct exports and simultaneously degrades the Rest of World supply channel by a stochastic fraction sampled from a commodity-specific Pareto distribution. The results diverge sharply across drug classes.

For cisplatin and carboplatin, the mean capacity loss is 16.8%, with a median of 16.0%. The 5th-percentile loss is 11.8% and the 95th-percentile loss is 24.8%. The probability of severe shortage (exceeding 30% capacity loss) is 0.01% — effectively zero. The platinum-based drugs survive the cascading shock because their upstream precursor supply chain (HS 284390) has minimal indirect Chinese exposure. South African mining operations, Swiss and German refining, and Belgian processing constitute the dominant supply corridor, and these pathways do not depend on Chinese starting materials to a significant degree.

For methotrexate, the mean capacity loss is 59.0%, with a median of 57.8%. The 5th-percentile loss is 51.3% and the 95th-percentile loss is 70.9%. The probability of severe shortage is 100%. Every single iteration — all 10,000 — breaches the 30% threshold. Methotrexate is catastrophically vulnerable to a cascading upstream shock because the heterocyclic compound supply chain (HS 293359) has an estimated 50–80% indirect exposure to Chinese starting materials. When China's direct exports are removed and the ROW channel is degraded by this fraction, the combined loss exceeds 50% of baseline capacity in every sampled scenario.

The divergence between 0.01% and 100% probability of severe shortage — for drugs analyzed within the same model, using the same data source, under the same disruption scenario — constitutes the central finding of this paper. It demonstrates that oncology drug supply chain risk is not uniform. It is commodity-specific, structurally determined by the upstream chemistry of each drug's precursor pathway, and invisible to any monitoring framework that treats "drug shortages" as a single undifferentiated category.

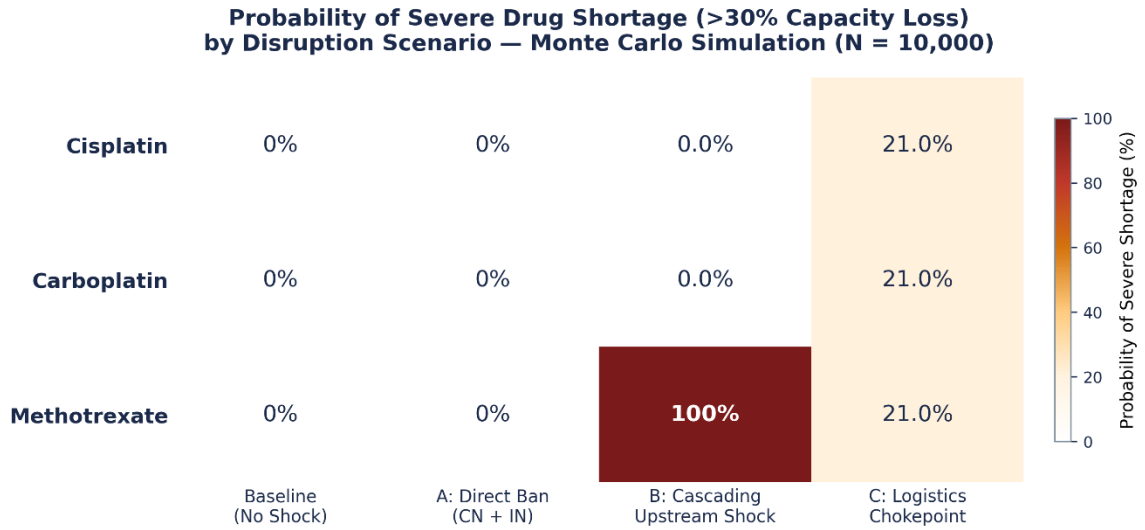

Figure 1. *Probability of Severe Drug Shortage (>30% Capacity Loss) by Disruption Scenario. Monte Carlo simulation, N = 10,000 iterations per scenario.*

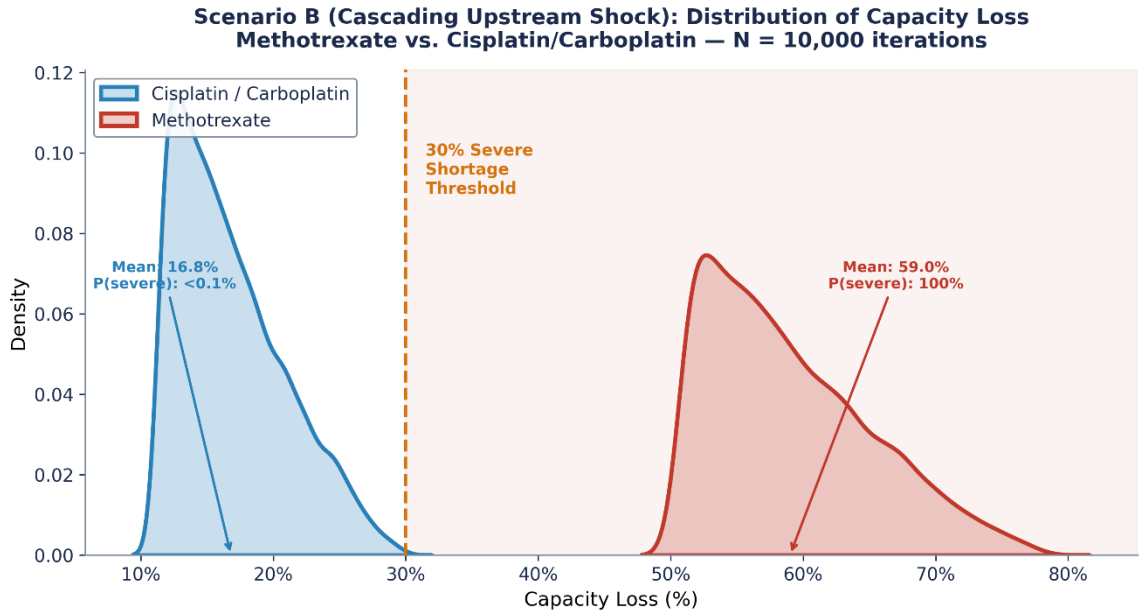

Figure 2. *Scenario B (Cascading Upstream Shock): Distribution of Capacity Loss. Methotrexate vs. Cisplatin/Carboplatin, with 30% Severe Shortage Threshold. N = 10,000 iterations.*

Figure 1 displays the probability of severe shortage across all four scenarios and all three drugs. The 100% cell for methotrexate under Scenario B is the dominant feature. Figure 2 overlays the capacity loss distributions for methotrexate and cisplatin under Scenario B, with a vertical reference line at the 30% severe shortage threshold. The methotrexate distribution sits entirely to the right of the threshold. The cisplatin distribution sits entirely to the left. There is no overlap.

3.2.4 Scenario C: Global Logistics Chokepoint

Under a systemic logistics disruption affecting all import corridors uniformly, all three drugs exhibit identical risk profiles. The mean capacity loss is 22.6%, with a median of 20.0%. The 5th-percentile loss is 9.0% and the 95th-percentile loss is 45.6%. The probability of severe shortage is 21.0% for all three drugs.

This result carries a distinct policy implication. The platinum-based drugs, which survive the geopolitical shock of Scenario B with minimal impact, are exposed to logistics disruption at a rate of one-in-five simulations. Methotrexate, which is already catastrophically vulnerable to geopolitical shock, faces an additional 21% probability of severe shortage from an independent logistics channel. The two risk vectors are additive, not substitutive. A supply chain that is resilient to one class of disruption is not necessarily resilient to another.

3.3 Summary of Risk Divergence

The simulation results partition the three drugs into two distinct risk classes.

Methotrexate faces a geopolitically driven vulnerability. Its precursor supply chain passes through chemical intermediates with heavy indirect Chinese dependency. A China-specific upstream shock produces a 100% probability of severe shortage. The policy response must target upstream chemical diversification — reducing the Rest of World supply chain's dependence on Chinese starting materials for heterocyclic compound synthesis.

Cisplatin and carboplatin face a logistically driven vulnerability. Their precursor supply chain is anchored in South African platinum mining, with minimal Chinese exposure. A China-specific shock barely registers. But a systemic logistics disruption — a port closure, a shipping corridor blockage, a pandemic-related capacity reduction — produces a 21% probability of severe shortage. The policy response must target logistics corridor redundancy, strategic buffer stockpiling, and domestic manufacturing capacity for platinum-based APIs.

A monolithic supply chain policy that treats all oncology drug shortages as a single problem will fail to address either risk class adequately. The SAPIR-Net model provides the commodity-specific resolution necessary to design targeted interventions.

4. Real-World Validation — The Intas Case Study

4.1 Overview

The predictive value of a supply chain risk model depends on its correspondence with observed events. The 2023 U.S. platinum chemotherapy shortage provides a natural experiment against which the SAPIR-Net model's outputs can be assessed. This section traces the causal chain of the shortage from regulatory trigger through patient impact and compares the observed supply reduction to the model's Scenario A predictions.

4.2 The Causal Chain

The sequence of events is fully documented in primary federal sources.

In November 2022, the FDA conducted an inspection of Intas Pharmaceuticals Limited at its sterile injectable manufacturing facility in Sanand, Gujarat, India (FEI 3004011473). The inspection identified five categories of current good manufacturing practice violations. These included systemic data integrity failures — quality control documents discovered destroyed in trash bags, an analyst observed pouring acid on analytical records, electronic batch records modified without audit trail review, and hundreds of chromatographic sequences aborted without adequate corrective action over a two-year period. The facility's environmental monitoring data for carboplatin injection batches were found to contain inaccurate colony forming unit counts. The FDA characterized the findings as reflecting an "ineffective quality system" with failures of "executive management oversight and control" and noted that similar violations had been cited in a prior 2019 inspection without adequate remediation (FDA, 2023).

On February 10, 2023, the FDA posted a cisplatin shortage to the CDER Drug Shortage Database. On April 28, 2023, a carboplatin shortage was posted. On June 1, 2023, the FDA placed the Intas facility on Import Alert 66-40, subjecting all products from the site to detention without physical examination at U.S. ports of entry. The formal warning letter was issued on July 28, 2023.

The supply impact was quantified by Reibel et al. (2025) in a retrospective cohort study using Flatiron Health electronic health record data from geographically diverse U.S. oncology clinics. Among 11,797 adults with advanced solid cancers initiating first-line therapy, platinum chemotherapy prescribing declined by 2.7 percentage points over the full shortage year (95% CI: −4.4% to −0.9%), corresponding to approximately 137 fewer patients than expected receiving platinum-based treatment. At peak shortage in June 2023, the decline reached 15.1 percentage points — from 72.9% of patients initiating platinum therapy in June 2022 to 57.8% in June 2023.

The FDA authorized emergency importation of cisplatin with non-U.S. labeling to mitigate the crisis. The cisplatin shortage was resolved in June 2024, though cisplatin reappeared on the ASHP shortage list in May 2025. As of April 2026, carboplatin remains in active shortage according to the FDA, with multiple manufacturers reporting unavailable or discontinued presentations. Methotrexate injection entered shortage in March 2023 and remains in active shortage as of April 2026 (FDA, 2026a; FDA, 2026b).

4.3 Model Comparison

The Intas event most closely corresponds to SAPIR-Net Scenario A (Direct Export Ban), which models the complete loss of direct bilateral imports from China and India. While the Intas shutdown was a single-facility quality failure rather than a bilateral trade disruption, its functional effect was equivalent: a major Indian production node was removed from the U.S. supply chain by regulatory enforcement.

Scenario A predicts a capacity loss of 2.95% for cisplatin and carboplatin. The observed full-year average reduction was 2.7% (95% CI: −4.4% to −0.9%). The model's point prediction falls within the observed confidence interval.

The peak reduction of 15.1% observed in June 2023 exceeds the Scenario A prediction. This is expected. Scenario A models a permanent, fully-adjusted state in which the market has had time to rebalance. The peak-month observation captures the acute phase of the disruption, before mitigation mechanisms (emergency imports, regimen substitution, inventory drawdown) take effect. The gap between the model's steady-state prediction and the acute-phase observation is consistent with the model's stated limitation of not capturing dynamic market adjustment (Section 2.6).

The USP Medicine Supply Map provides additional corroboration. As of January 2025, cisplatin carried a Vulnerability Score of 79.1% — nearly four times the average prescription drug score of approximately 20% (Malta et al., 2025). The facility-level data confirm the concentration mechanism: one Indian facility produced 24% of U.S. cisplatin supply and 8% of U.S. carboplatin supply, while simultaneously manufacturing 84% of temsirolimus and 43% of carmustine. This multi-drug production concentration at a single site is the real-world instantiation of the shared-node vulnerability that the SAPIR-Net graph structure is designed to capture.

4.4 Implications for the Cascading Shock Hypothesis

The 2023 event was a quality-driven single-facility failure. It did not involve a geopolitical disruption of Chinese upstream supply. Scenario B — the cascading upstream shock in which Chinese starting material supply is disrupted and the Rest of World channel is probabilistically degraded — has not yet been observed in the historical record for platinum-based oncology drugs.

This absence is itself informative. The 2023 shortage was triggered by the mildest class of supply chain disruption in the SAPIR-Net framework: the loss of one country's direct exports (functionally, a subset of Scenario A). It produced a 15.1% peak prescribing reduction, emergency federal intervention, and a carboplatin shortage that has persisted for three years. If a single-facility quality failure in India generates this level of impact, a systemic disruption of Chinese upstream chemical supply — which would simultaneously degrade the European, Japanese, and Indian production channels that constitute the bulk of Rest of World exports — would produce consequences of a qualitatively different magnitude.

For methotrexate, where the model predicts 100% probability of severe shortage under Scenario B, the absence of a historical test case does not provide reassurance. It provides a window for preemptive intervention.

5. Policy Implications and Strategic Recommendations

5.1 The Failure of Monolithic Reshoring

The prevailing policy discourse on pharmaceutical supply chain resilience centers on "reshoring" — relocating API and finished dosage form manufacturing to the United States or to allied nations. Executive Order 14017 and subsequent legislative proposals have directed federal investment toward expanding domestic production capacity for critical drugs and materials.

Reshoring is a necessary component of long-term supply chain resilience. It is not, by itself, sufficient. The SAPIR-Net results demonstrate that the upstream risk structure of oncology drug supply chains is commodity-specific, and a single policy instrument applied uniformly across drug classes will misallocate resources.

Reshoring platinum-based API manufacturing to the United States would address the logistics vulnerability identified in Scenario C (21% probability of severe shortage). It would have minimal impact on the geopolitical vulnerability identified in Scenario B, because the platinum supply chain's low indirect Chinese exposure means a China-specific shock produces negligible impact on these drugs regardless of where the final API synthesis occurs. Conversely, reshoring methotrexate API manufacturing would be ineffective if the domestic facility continued to source heterocyclic intermediates from the same Chinese-dependent global supply chain. The vulnerability would relocate, not resolve.

Effective policy must differentiate between the drug classes and target interventions to the specific risk vector each class faces.

5.2 Recommendations for Heterocyclic Compounds (Methotrexate)

The 100% severe shortage probability for methotrexate under cascading upstream shock is driven by a single variable: the 50–80% indirect Chinese dependency in the heterocyclic compound supply chain. Reducing this variable is the only mechanism that materially decreases methotrexate's vulnerability.

Recommended interventions include investment in non-Chinese starting material sources for heterocyclic pharmaceutical intermediates, with priority given to European, North American, and allied Asian (Japan, South Korea) chemical manufacturers. Federal procurement incentives under the Defense Production Act or successor authorities should condition price preferences on demonstrated supply chain diversification — specifically, on the fraction of starting materials sourced from non-Chinese suppliers. The NASEM Recommendation 1 framework — requiring public disclosure of API and starting material sourcing on pharmaceutical labels — would provide the transparency necessary for purchasers to evaluate and reward diversified supply chains.

5.3 Recommendations for Platinum Compounds (Cisplatin, Carboplatin)

The platinum-based drugs face a structurally different risk profile. Their upstream precursor supply is geopolitically secure — South Africa, the dominant source, maintains independent mining and refining operations with minimal Chinese exposure. The vulnerability is logistical: a systemic disruption to global shipping corridors or port infrastructure produces a 21% probability of severe shortage regardless of the geopolitical origin of the disruption.

Recommended interventions include expansion of the Strategic National Stockpile to include cisplatin and carboplatin injectable formulations with appropriate cold-chain management. The current stockpile framework, designed primarily for chemical, biological, radiological, and nuclear countermeasures, does not systematically address chronic oncology drug supply vulnerability. A rotating buffer stock of 90–180 days of national consumption, managed in coordination with FDA shortage monitoring, would attenuate the acute-phase prescribing disruption observed during the 2023 shortage. Concurrently, the FDA should formalize expedited regulatory pathways for emergency authorization of alternative suppliers when primary production nodes fail, building on the temporary importation mechanism used during the 2023 cisplatin crisis.

5.4 Recommendations for Federal Monitoring Infrastructure

The 6-Digit Dilution Effect identified in this analysis represents a structural limitation of current trade-data-based supply chain monitoring. The Harmonized System classification at the 6-digit level cannot distinguish pharmaceutical-grade precursor flows from bulk industrial commodity volumes. Federal agencies relying on Comtrade or similar trade databases for pharmaceutical supply chain risk assessment will systematically underestimate or misattribute concentration risk unless they supplement trade data with facility-level production information.

The SAPIR-Net model demonstrates that a three-layer network approach — combining commodity-level trade data, chemical dependency mappings, and parameterized indirect exposure estimates — can produce actionable risk differentiation even within the constraints of publicly available data. The model's alignment with the observed 2023 shortage impact suggests that this approach has predictive validity. Federal adoption of network-based supply chain models, integrated with the facility-level production and API sourcing data called for in NASEM Recommendation 1 and the White House E.O. 14017 review, would substantially improve the resolution and accuracy of pharmaceutical infrastructure risk assessment.

Upstream supply chain monitoring alone, however, addresses only one dimension of the vulnerability. The Intas case study (Section 4) demonstrates that facility-level equipment and process control failures are an independent and empirically observed trigger of national drug shortages. A complete resilience architecture requires monitoring at multiple levels of abstraction: the macro-geopolitical commodity layer addressed by SAPIR-Net, and the micro-level facility asset health layer where equipment degradation patterns can be detected and acted upon before they produce regulatory enforcement and supply disruption. Complementary frameworks integrating real-time equipment telemetry with national shortage surveillance — such as hardware-agnostic, GxP-compliant predictive maintenance platforms designed for pharmaceutical manufacturing

environments (see Kazmi, 2026) — would close the gap between upstream risk identification and facility-level failure prevention. The convergence of macro-level supply chain intelligence and facility-level predictive maintenance constitutes the minimum viable monitoring architecture for a pharmaceutical infrastructure that the United States cannot afford to lose.

6. Conclusion

SAPIR-Net is a macro-geopolitical dependency graph model that maps the upstream chemical precursor supply chains of three essential U.S. oncology drugs — cisplatin, carboplatin, and methotrexate — and quantifies their vulnerability to geopolitical and logistical disruption through Monte Carlo simulation.

The model identifies a sharp, structurally determined divergence in risk profiles. Methotrexate is catastrophically exposed to cascading geopolitical shock, with a 100% probability of severe shortage when Chinese upstream chemical supply is disrupted, driven by the heterocyclic compound supply chain's 50–80% indirect dependency on Chinese starting materials. Cisplatin and carboplatin are resilient to the same geopolitical shock but face a 21% probability of severe shortage under systemic logistics disruption. These are different diseases of the same supply chain. They require different treatments.

The model's Scenario A output (2.95% capacity loss for platinum-based drugs) aligns with the observed 2.7% average prescribing reduction during the 2023 platinum chemotherapy shortage, providing partial empirical validation. The complete causal chain of that shortage — from a single Indian facility's manufacturing quality failures through FDA enforcement action to a three-year ongoing carboplatin shortage — confirms the single-point-of-failure mechanism that the SAPIR-Net graph structure is designed to detect.

Two structural findings have immediate policy relevance. First, the 6-Digit Dilution Effect renders standard trade-data-based monitoring blind to the pharmaceutical-specific concentration risk embedded within broad commodity classifications. Federal supply chain surveillance must supplement trade data with facility-level production and sourcing information to achieve actionable resolution. Second, the indirect dependency channel — in which Rest of World suppliers themselves depend on Chinese starting materials — creates a hidden transmission mechanism that bilateral trade statistics do not capture. Risk models that assess only direct country-to-country trade flows will systematically understate the true exposure.

The framework is extensible. The three-layer graph architecture and modular simulation engine can accommodate additional drugs, additional HS codes, additional geopolitical source nodes, and higher-resolution facility-level data as it becomes available through the transparency mechanisms recommended by NASEM and the White House. The model is designed not as a static assessment but as a monitoring instrument — one that can be updated as trade patterns shift, new shortage events occur, and federal data infrastructure improves.

The supply chains that deliver cancer drugs to American patients are long, concentrated, and partially opaque. SAPIR-Net provides a means to see through that opacity, quantify the risk it conceals, and direct limited policy resources toward the specific vulnerabilities that matter most.

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